Week 4: Protocol Pro

This project involves the design of a quadcopter drone system, with a device connection diagram created using draw.io.

The Holybro Pixhawk 6C flight controller—featuring an integrated IMU and barometer—was selected for this build. A Holybro M10 GPS and compass module is connected externally via the GPS1 port: the GPS utilizes UART, while the compass uses I²C. A FrSky X8R receiver transmits remote control commands to the flight controller via SBUS.

The power distribution path is as follows: Battery → PM02 Power Module → Power Distribution Board (PDB) → Four Electronic Speed Controllers (ESCs). The PM02 module powers the flight controller and provides battery telemetry data. The flight controller manages the ESCs using four independent PWM signals, with each ESC driving a single motor.

This design utilizes PX4 firmware. The provided image is a simplified functional connection diagram; not all individual wires are shown in detail. As the system has not yet been assembled or tested, actual wiring requires verification of pinouts, voltage levels, and device compatibility.

References:
https://docs.holybro.com/autopilot/pixhawk-6c
https://holybro.com/products/m10-gps
https://holybro.com/products/pm02-v3-12s-power-module